

Katarina Andrade

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EDUCATION

Northwestern University, Evanston, IL

Expected: Jun 2027

Bachelor of Science in Communication

Majors in Theatre & Physics (Astronomy concentration)

GPA: 3.86/4.0

Relevant coursework: Creating Inclusive Science; Observational Astrophysics; Extragalactic Astrophysics and Cosmology; Science Communication

Additional Training: Granville Academy (Yale, Summer 2025) – Curriculum blending astrophysics tutorials with professional development to analyze systemic barriers and cultivate inclusive research environments.

RESEARCH EXPERIENCE

BAT AGN Spectroscopic Survey (BASS Collaboration)

Sep 2025 – Present

Research Assistant

Advisor: Dr. Alessandro Peca

Quantifying the percentage of local obscured Active Galactic Nuclei (AGN) displaying [Ne V] emission lines to calibrate benchmarks for interpreting the nature of high-redshift ($z > 4$) Little Red Dots (LRDs) with JWST observations. [BASS Survey link available here.](#)

Active Galactic Nuclei Group, Yale University

Jun 2025 – Sep 2025

Research Assistant

Advisor: Dr. Meg Urry

Analyzed JWST/NIRSpec spectra of high-redshift galaxies ($z > 4$) to identify AGN signatures using [Ne V] emission; built a pipeline for source selection based on color, compactness, and emission-line detectability. [Write up accessible here.](#)

Observational Galaxy Evolution Group, Northwestern University

Jun 2024 – Jun 2025

Research Assistant

Advisor: Dr. Allison Strom

Investigated the reliability of spectral stacking methods to reveal faint emission lines in ~ 50 high-redshift ($z \approx 2-3$) galaxies; analyzed Keck/KBSS and JWST CECILIA (Chemical Evolution Constrained using Ionized Lines in Interstellar Aurorae) survey data. Project [Grant Proposal](#) and [Conference Poster](#) available online.

Northwestern School of Communication's Molosky Summer Award

2026

Northwestern's Summer Internship Grant Program (SIGP)

2025

Alumnae of Northwestern University Undergraduate Research Scholar

2024

Northwestern Academic Year Undergraduate Research Grant (AYURG)

2024

Northwestern Summer Undergraduate Research Grant (SURG)

2024

PRESENTATIONS

Talk, Summer Research Symposium, "A Spectroscopic Analysis of the Little Red Dots"

2025

Yale Department of Astronomy, USA

Poster, Northwestern Undergraduate Research Expo, "Let Them Shine: Revealing Faint Galaxies with the Spectral Stacking Technique"

2025

Northwestern University, USA

Talk, Observational Galaxy Evolution Group Meeting, "Analyzing faint emission lines in distant galaxies through stacks"

2025

Northwestern University, USA

OUTREACH EXPERIENCE

REACH: Research Experiences in Astronomy at CIERA for High School Students *Mar 2026 – Sep 2026*

Instructional Support Mentor

Assisted high school students through coding training, educational activities, and telescope viewing at Dearborn Observatory; mentored a student through a dedicated three-week independent astrophysics project.

Wright Lab, Yale University

Jul 2025

Science Outreach Volunteer

Led hands-on acoustics and thermodynamics science activities for the Pathways to Science Summer Scholars program for high school students.

WiP (We're in Physics), Northwestern University

May 2024 – Present

Co-President

Lead undergraduate club supporting women and gender minorities in physics; coordinated conference attendance funding, peer research placement workshops, and structured mentorship programs.

Dearborn Observatory, Northwestern University

Nov 2023 – Present

Host

Host public observing sessions by operating the historic 18.5" refracting telescope; coordinate Public Night tours and solar observing sessions.

Sciety Lab Institute, Brazil

Jan 2022 – Dec 2023

Co-founder and Coordinator

Co-founded a remote non-profit engaging female-identifying students through the Women and Girls in Galaxies program; organized collaborative galaxy morphology classification campaigns utilizing the NAOJ GALAXY CRUISE citizen science platform, hosted panels with astronomers, and ran scientific writing workshops.

TEACHING EXPERIENCE

Northwestern Center for Talent Development

Jan 2026 – Present

Teaching Assistant

Instruct STEM learning for Pre-K and Grades 1–2 students in enrichment courses including Treasure Hunt: Math and Maps and Kids Chemists; Guided hands-on experiments, adapting scientific concepts into age-appropriate lessons and incorporating accommodations for students with learning disabilities.

Perez Elementary School

Jan 2024 – Apr 2024

Classroom Volunteer

Collaborated with lead Pre-K teachers to facilitate scientific and performing arts activities.

SKILLS & INTERESTS

Technical Skills	Python, PyQSOfit, GitHub, telescope instrumentation, science communication
Languages	Portuguese (native), English (fluent), Spanish (advanced), French (beginner)
Interests	Evolution of black holes and galaxies, Little Red Dots, Active Galactic Nuclei (AGN), Galaxy Evolution, Emission-line diagnostics, Women and FGLI inclusion in STEM